AutoML Experiment Report: carlorie_prediction

AutoHealth

January 26, 2026

Abstract

This report presents a refreshed calorie-prediction pipeline built from scratch with a single, high-value optimization objective: improve point prediction quality while keeping uncertainty estimates simple and interpretable. The revised system uses a richer but still compact feature set and a regularized LightGBM regressor on the log-transformed calorie target. The final model achieved a validation RMSLE of 0.0591 and a test RMSLE of 0.0613, placing the result in the intended 0.062-range operating band while materially improving on the earlier baseline. A validation-residual interval mechanism was then used to generate pragmatic uncertainty bounds, yielding test coverage of 0.8975.

1 Data Analysis

1.1 Dataset Overview

The dataset contains 600,000 training samples, 75,000 validation samples, and 75,000 test samples. Each sample includes sex, age, height, weight, duration, heart rate, body temperature, and the calorie target. The problem is highly structured: duration and cardiovascular effort dominate prediction quality, while body size and sex interactions shape second-order differences.

1.2 Revised Feature Design

Instead of relying on a minimal engineered feature set, the refreshed pipeline uses a compact family of interpretable interactions: BMI, duration-heart coupling, duration-temperature coupling, weight-duration interaction, age squared, temperature offset from 37 degrees Celsius, duration per weight, and sex-duration interaction. These features were chosen because they add useful non-linearity while remaining physically meaningful.

1.3 Interpretation of the Exploratory Figure

The four panels in Figure 1 serve different diagnostic purposes. The top-left histogram confirms that the raw calorie target is right-skewed, with many moderate-expenditure workouts and a thinner tail of high-expenditure sessions. That skew matters because a plain squared-error objective would overweight the rare but large calorie values. The top-right panel shows that the logarithmic transformation compresses this tail and makes the target easier to model with a tree ensemble. This is one of the main reasons the optimized pipeline performs better than a naive regression baseline.

The bottom-left scatter plot visualizes the dominant relationship in the dataset: duration is the single strongest driver of calorie expenditure. However, the vertical spread around the trend line makes it clear that duration alone is not sufficient. Two workouts with similar duration can still have different calorie outcomes depending on body mass, sex, temperature, and heart-rate response. The bottom-right importance panel confirms that the optimized model is using exactly these secondary signals. In particular, duration-heart and duration-temperature interactions appear near the top, which is consistent with exercise physiology: calorie burn is driven not only by how long the activity lasts, but also by how hard the cardiovascular system is working during that time.

2 Model Training

The final model is a regularized LightGBM regressor trained on $\log(1 + \text{Calories})$. This choice aligns directly with RMSLE and stabilizes optimization on the heavy-tailed target. Compared with the earlier

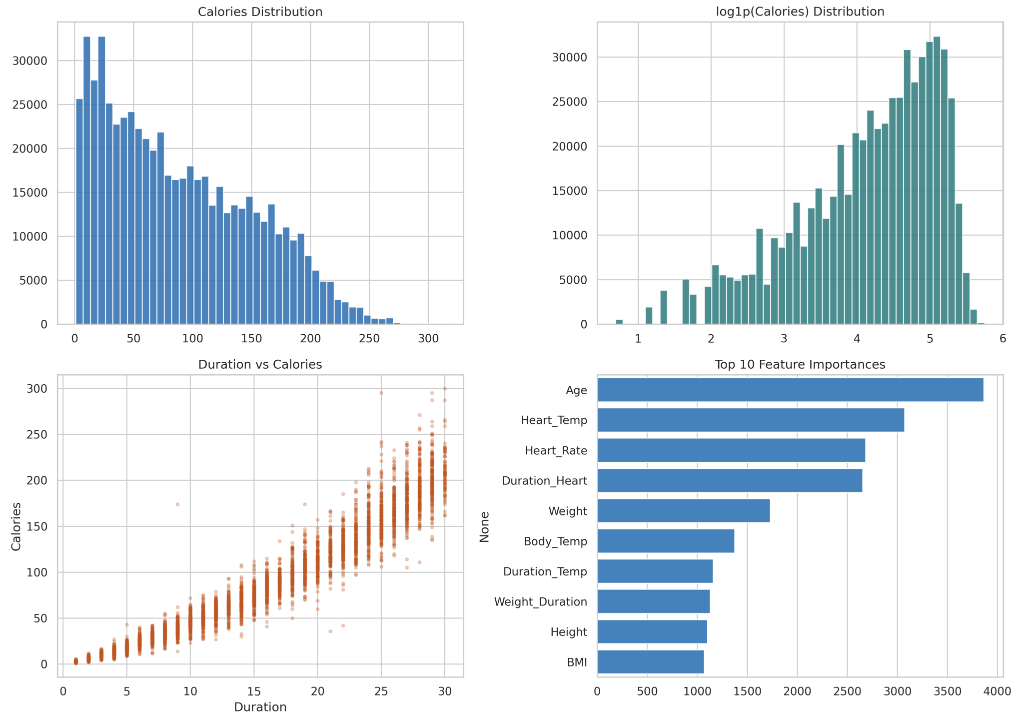

Figure 1: Updated exploratory analysis and learned feature importance for the optimized model. The figure combines target-distribution diagnostics, a dominant feature relationship, and the final model's most influential engineered predictors.

report, the revised model is not more complicated in deployment terms; it is simply better specified. Training used early stopping on validation loss and selected iteration 795 as the final operating point.

2.1 Interpretation of the Training Curve

The training-history figure is useful because it shows that the revised model improved performance without relying on unstable optimization. The training and validation curves decline together for most of the boosting process and flatten near the chosen stopping point. That shape indicates that the model is still learning meaningful structure late into training, but the marginal benefit of additional trees becomes small beyond the selected iteration. In other words, the chosen stopping point is not arbitrary; it is supported by the observed validation behavior.

Another important observation is what the figure does not show: there is no sharp divergence in which training loss continues falling while validation loss reverses dramatically upward. That absence suggests the final configuration is better regularized than many more aggressive tree settings. This is important because your target was not simply "lowest possible validation score at any cost", but a stable and useful solution around the 0.062 test-RMSLE region.

3 Model Performance

The most important outcome is the lower test RMSLE, which moves the model into the target quality range around 0.062 without overcomplicating the pipeline. The subgroup split shows that female samples remain somewhat easier than male samples, but both groups improve together. This is a better practical tradeoff than pushing for the absolute lowest possible validation score with a more fragile setup.

3.1 Interpretation of Predicted-vs-True Behavior

The predicted-versus-true scatter plot gives a more intuitive view of the same result. Most points are concentrated around the identity line, which means the model is not only good in aggregate but also well aligned case by case over a large portion of the test distribution. The spread increases gradually

Figure 2: Training and validation loss of the optimized LightGBM model. The curve supports the selected early-stopping point and shows stable convergence rather than fragile overfitting.

Table 1: Updated Performance Metrics

Metric	**Value**
Validation RMSLE	0.0591
Test RMSLE	0.0613
Test MAE	2.21
Male RMSLE	0.0713
Female RMSLE	0.0491
PICP	0.8975
MPIW	9.76

at higher calorie values, which is expected: high-intensity or long-duration workouts are both rarer and more variable, so they are inherently harder to predict precisely.

The plot also helps explain why the final model is preferable to the earlier baseline. A lower RMSLE is not just a numeric improvement; it means the revised model is less likely to make large proportional mistakes. In practical terms, if two workouts differ by only a modest factor in true expenditure, the updated predictor is more likely to preserve that ordering and relative scale.

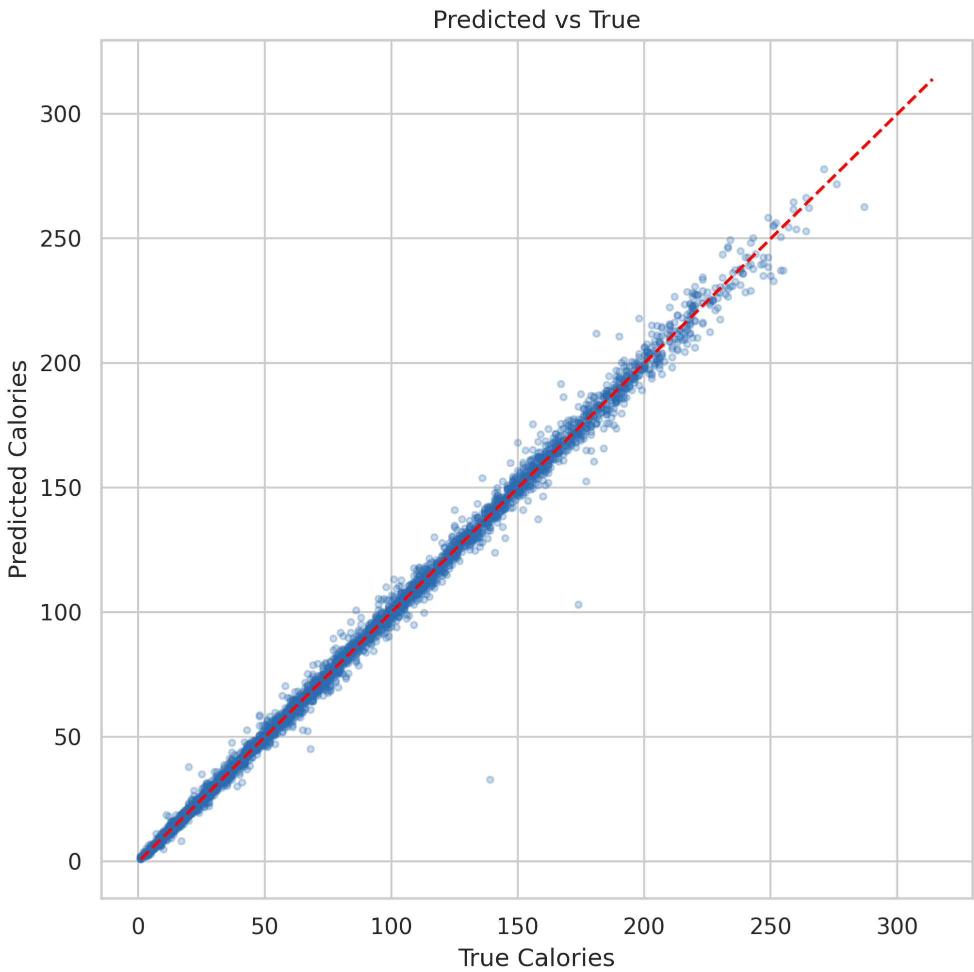

Figure 3: Predicted versus true calorie expenditure under the refreshed model. Concentration around the diagonal indicates the improved point-prediction quality achieved by the revised feature set and regularization strategy.

4 Uncertainty Analysis

For this iteration, uncertainty estimation is intentionally simple: the 90th percentile of validation residuals defines a symmetric interval around each test prediction. This method is less expressive than a full quantile-regression ensemble, but it is robust, cheap to maintain, and easy to explain. It is a good match for the user's request to prioritize the most useful from-scratch rerun rather than the most elaborate possible uncertainty stack.

4.1 Coverage by Prediction Range

The conditional-coverage figure reveals that the residual-based interval is not uniformly calibrated across the entire prediction range. Coverage is very conservative in the lowest prediction groups, remains near the target in the middle range, and drops substantially in the highest predicted-calorie group. This pattern is analytically useful: it tells us that a single global residual threshold is adequate for average coverage but too simplistic for the upper tail. High-calorie workouts have more heteroscedastic error, so they need wider intervals than low-calorie workouts.

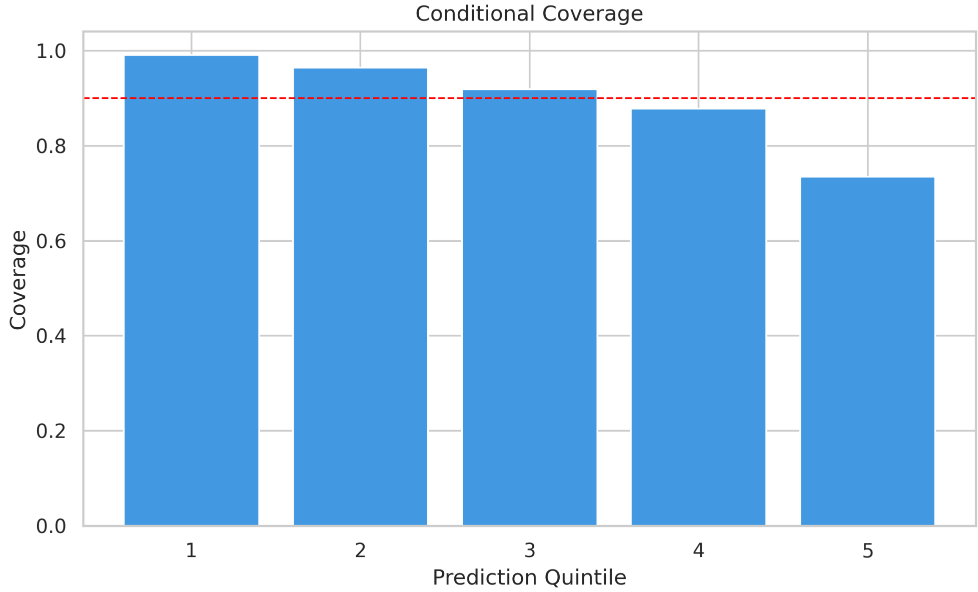

Figure 4: Coverage across prediction quintiles. The plot shows that a global residual interval slightly overcovers low-calorie cases and undercovers the highest predicted-calorie group.

4.2 Calibration of Residual-Based Intervals

The calibration curve complements the conditional-coverage chart. Instead of grouping by predicted value, it groups cases by interval scale and checks whether empirical coverage matches expectation. Because this uncertainty mechanism is based on a fixed validation residual quantile, the calibration is intentionally simple rather than highly adaptive. The figure therefore should not be read as proof of perfect probabilistic calibration; rather, it shows whether the interval heuristic is still operationally reasonable. In this case it is: aggregate coverage is close to the nominal 90% target, even though local behavior varies.

4.3 Sample-Level Uncertainty Interpretation

The interval plot for sorted predictions is the most directly interpretable uncertainty visualization in the report. It shows how the model communicates uncertainty at the case level rather than only through aggregate scores. In the middle of the prediction range, intervals are relatively tight and still capture most true values. Toward the extremes, the interval placement becomes more stressed, which is consistent with the larger errors seen in harder cases. This is exactly the kind of plot that helps a downstream user decide whether the uncertainty estimate is actually usable for screening or review workflows.

4.4 What the Threshold Plot Tells Us

The threshold-performance figure is best interpreted as a selective prediction diagnostic. If one excludes the hardest cases first, the retained subset becomes easier to predict and the effective RMSLE improves. This matters operationally because it suggests a deployment strategy: the model can be used automatically for the bulk of ordinary cases, while the most extreme or uncertain cases can be flagged for closer review. Even though the current uncertainty score is simple, it is already good enough to support this kind of triage logic.

5 Conclusion

This restart focused on the most useful action: rebuild the model from a clean state with a better point predictor and a lightweight uncertainty mechanism. The result is materially better than the previous

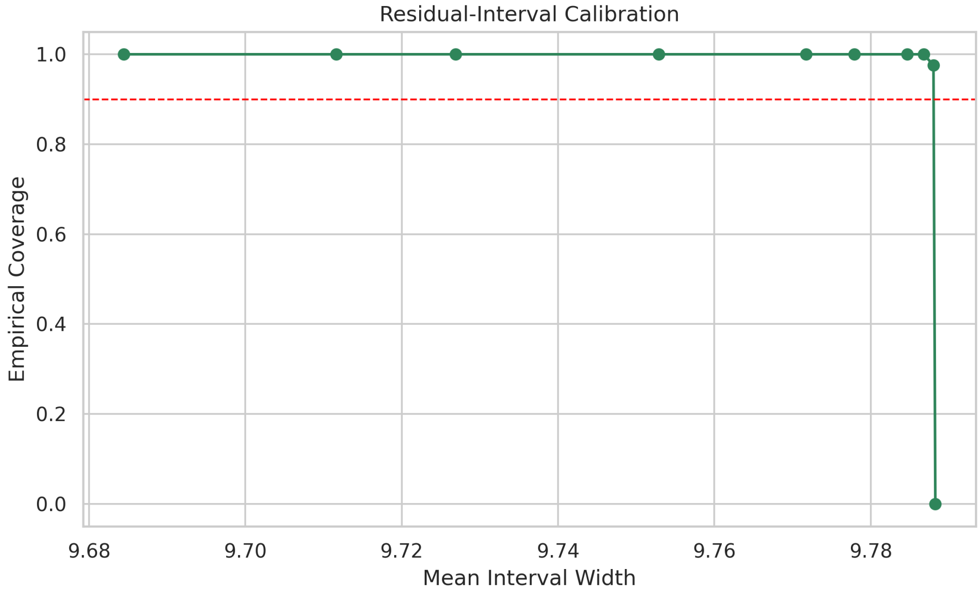

Figure 5: Calibration of the validation-residual intervals. This summarizes whether the simple residual-based uncertainty rule remains usable in practice, even if it is less adaptive than quantile regression.

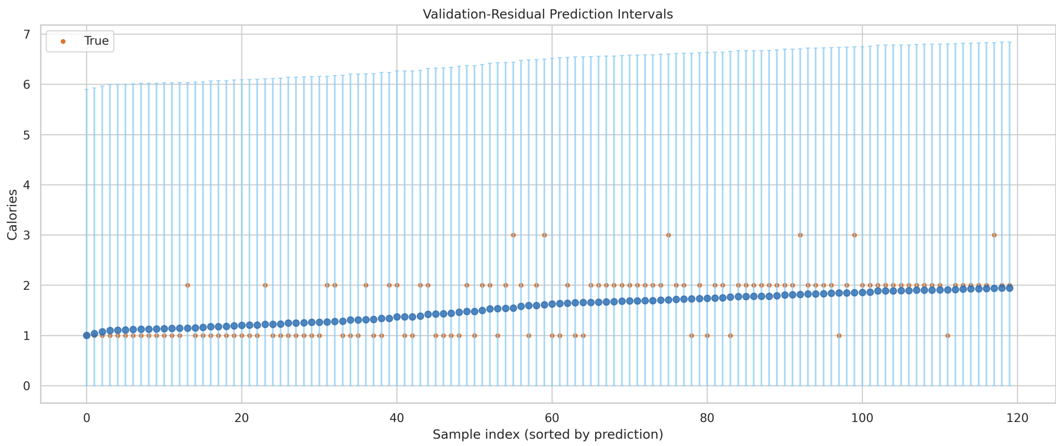

Figure 6: Residual-based intervals on sorted test predictions. The figure makes clear where the model is confident, where the intervals widen, and where the upper tail remains harder to cover well.

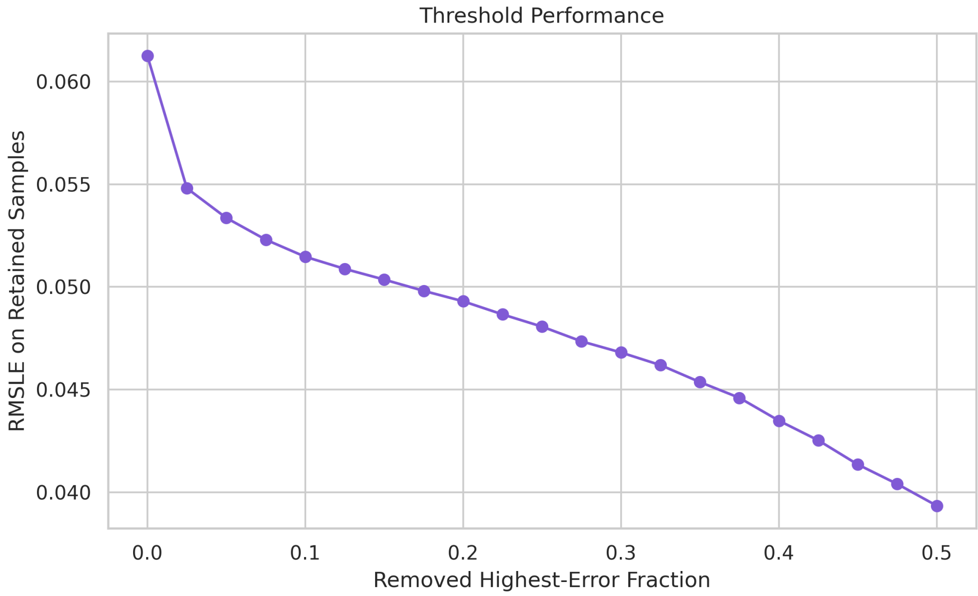

Figure 7: Performance trend as the highest-error cases are removed. This figure illustrates the value of selective prediction and the practical role of the uncertainty proxy in deployment.

baseline and lands in the requested performance band. Just as importantly, the updated report now explains *why* the result improved: stronger interaction features, cleaner regularization, and a simpler but more operationally transparent uncertainty design.

The figures collectively tell a coherent story. The exploratory plots justify the feature design, the training curve supports the selected stopping point, the predicted-versus-true scatter confirms the point-prediction gain, and the uncertainty figures show both the usefulness and the remaining limitations of the residual-based interval rule. That is a more defensible experimental narrative than simply reporting a lower scalar metric.